Generative AI as a Feedback Partner in Higher Education: A PRISMA 2020 Systematic Review and Meta-analysis on Learning and Performance

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Abstract

Background: Generative AI (GenAI) and large language models (LLMs) are rapidly being embedded in higher education as *feedback partners* within formative assessment workflows. Beyond efficiency gains, their pedagogical value depends on how such systems mediate learners' cognitive and self-regulatory processes and on whether observed performance gains reflect durable learning rather than superficial output optimization. **Objective:** This study synthesizes empirical evidence on the effects of GenAI-supported feedback on learning and performance in higher education, and examines instructional and evaluative moderators related to task type, feedback design (rubric anchoring, worked examples/counterexamples, actionability with verification), support for self-regulated learning (SRL), and process-sensitive instrumentation (learning analytics, trace indicators, and neuro/biometric measures). **Methods:** Following PRISMA 2020 guidelines, we conducted a systematic review of peer-reviewed journal articles and conference papers indexed in the Web of Science Core Collection and Scopus (2022–2025). Two reviewers independently screened records and extracted data. When sufficient information was available, standardized mean differences (Hedges' $\boldsymbol{g}$) were computed and synthesized using random-effects meta-analysis. To ensure independence, the primary synthesis used *one consolidated product effect per study*, yielding 31 *study-level* effects. Pre-specified subgroup analyses

and *exploratory*, parsimonious meta-regressions examined design-related moderators with caution due to limited power. Publication bias was explored using funnel plots, Egger-type regression, and trim-and-fill sensitivity analyses when appropriate; additional sensitivity analyses examined design subsets. **Results:** Across $\boldsymbol{k = 31}$ pooled study-level effects, GenAI-supported feedback demonstrated a positive pooled effect on learning and performance ($\boldsymbol{g = 0.541}$, 95% CI [0.460, 0.623]), with moderate heterogeneity ($\boldsymbol{I^2 = 34.76\%}$). Larger effects were consistently associated with instructionally integrated designs in which feedback was rubric-anchored, explicitly actionable, and embedded in iterative revision and verification cycles. Evidence on SRL and process outcomes was less frequently reported but was stronger in studies employing trace-based indicators (e.g., revision iterations, feedback-to-revision latency, retry patterns), which also mitigate construct-validity threats from *performance inflation* (product gains without demonstrable process change). Neuro/biometric measures appeared in a small subset of studies and primarily contributed to mechanism-oriented explanations rather than effectiveness estimation at scale. **Conclusions:** GenAI can meaningfully enhance feedback processes in higher education when designed as a structured, criterion-referenced feedback partner that supports iterative learning and self-regulation. Effectiveness is contingent on pedagogical constraints such as rubric anchoring, exemplification, and verifiability, as well as on the inclusion of process-oriented evaluation. Scalable log- and trace-based analytics offer a practical pathway for SRL-aligned assessment, while neuro/biometric approaches represent an emerging direction for theory-driven mechanism research.

Keywords: generative AI; large language models; feedback; formative assessment; higher education; systematic review; meta-analysis; PRISMA 2020; self-regulated learning; learning analytics

1 Introduction

Generative artificial intelligence (GenAI), and in particular large language models (LLMs), has rapidly transitioned from experimental deployment to routine use across higher education [1–3]. This shift is especially visible in learning activities that rely on iterative drafting and revision—including programming assignments, academic writing, design briefs, laboratory reports, and project-based deliverables—where students increasingly engage LLMs to obtain explanations, suggestions, and feedback. For instructors and institutions, the potential benefits are substantial: reduced feedback latency, scalable formative assessment, and more continuous instructional support for increasingly heterogeneous cohorts [1, 2]. At the same time, the very properties that make LLMs attractive—linguistic fluency, broad coverage, and conversational adaptability—also raise persistent pedagogical, ethical, and validity concerns: feedback quality can be inconsistent, difficult to verify, and prone to drift from intended learning outcomes, while students may rely on assistance in ways that obscure whether observed improvements reflect deeper learning or superficial output refinement [2, 4].

Feedback is among the most powerful instructional practices in higher education, yet its effectiveness depends on conditions that are difficult to sustain at scale,

including timeliness, specificity, alignment with assessment criteria, and learners' capacity to interpret and act on guidance [5, 6]. Contemporary accounts further emphasize feedback as a dialogic and relational process in which uptake, judgement of quality, and subsequent revision actions are central mechanisms [7–9]. GenAI can alter this ecology by enabling interactive, on-demand feedback dialogues; however, these affordances do not constitute a pedagogical method in themselves. Without explicit instructional-design constraints—such as rubric anchoring, transparent criteria, worked examples/counterexamples, and routines for verification—GenAI-supported feedback may remain poorly calibrated to course standards and weakly aligned with intended learning outcomes [6, 9]. Accordingly, we treat GenAI not as a feedback method, but as a *feedback delivery substrate* whose educational value is determined by how it is designed, constrained, scaffolded, and evaluated within an assessment workflow.

A central limitation of the empirical literature is its predominant focus on *product* outcomes (e.g., grades, rubric scores, test performance), accompanied by comparatively sparse measurement of *process* outcomes that reflect self-regulated learning (SRL). SRL frameworks emphasize cyclical regulation across planning, performance/-monitoring, verification, and revision—processes that are critical for durable learning and transfer beyond a single task [10–12]. This creates a construct-validity challenge in AI-mediated assessment: *product gains without process evidence may reflect output optimization rather than durable learning.* If GenAI primarily improves surface features of student work (e.g., wording, formatting, minor code fixes) without strengthening underlying self-regulatory processes, observed gains may not generalize. This risk is particularly salient in formally constrained domains such as programming, where LLM-generated solutions can be *fluent but functionally wrong* (a "fluency trap"): code may appear plausible and compile under common cases yet fail edge conditions, hidden tests, or robustness requirements. When learners accept fluent outputs without verification, performance may stagnate or even decline despite increased perceived progress [4]. Conversely, when GenAI is embedded in structured workflows that require learners to plan, justify revisions, and *verify* changes (e.g., unit tests, reproducible runs, static analysis, checklist-based checks), it may function as an accelerator of feedback cycles that supports both SRL and skill development [9, 12].

From a measurement perspective, GenAI-supported feedback environments also create a practical opportunity: even minimal instrumentation can yield scalable indicators of process change. Interaction logs and learning analytics can capture iteration counts, feedback-to-revision latency, retries/resubmissions, and traces of planning and checking activities (e.g., checklists, unit tests, self-explanations, or evidence-linked revisions). Such trace indicators complement self-report and help discriminate "better answers" from "better learning" [13–16]. Neuro- and biometric measures (e.g., EEG, eye tracking, fNIRS, cognitive-load proxies) can provide mechanistic insight into attention or workload dynamics, but are typically feasible only in targeted subsamples; they are best positioned as explanatory evidence rather than cohort-level evaluation requirements. In the current evidence base, neuro/biometric studies remain relatively scarce and heterogeneous, underscoring the need for cautious interpretation and for integration with trace-based process measures when mechanism claims are advanced.

Conceptually, we frame GenAI-supported feedback as a sociotechnical intervention embedded within an assessment and learning workflow. Let Y denote a product outcome (e.g., rubric-based performance), P a process construct reflecting SRL-consistent behaviors, D instructional design constraints (e.g., rubrics, examples/counterexamples, actionability with verification), and Z the quality of instrumentation (e.g., trace availability). A simplified mediation perspective is:

$$\text{GenAI} \rightarrow (D, Z) \rightarrow P \rightarrow Y, \tag{1}$$

where D and Z are implementable design and measurement choices that may determine whether GenAI-supported feedback translates into observable SRL process change and, subsequently, learning gains. Importantly, this framing motivates actionable moderator analyses (design and evaluability) while avoiding strong causal claims that the aggregate evidence cannot support.

Despite rapidly growing interest in GenAI for education, empirical findings remain dispersed across disciplines, task types, and study designs, and vary widely in how GenAI-supported feedback is operationalized and evaluated. Recent reviews and meta-analyses report positive average effects but substantial heterogeneity, reinforcing the need to identify boundary conditions tied to task demands, feedback-design constraints, verification routines, and process-sensitive evaluation [17–19]. There is therefore a need for a PRISMA 2020-compliant systematic review and meta-analysis that not only quantifies effects where possible, but also clarifies *when* and *why* GenAI-supported feedback is more likely to support learning rather than superficial output optimization—and, in domains such as programming, when verification is a necessary design condition to avoid the fluency trap.

1.1 Research questions and contributions

Guided by this framework, the review addresses three research questions (RQs):

- **RQ1 (Effectiveness):** What is the overall effect of GenAI-supported feedback on learning-related *product* outcomes in higher education?
- **RQ2 (Process/SRL):** How does GenAI-supported feedback relate to *process* outcomes, particularly SRL-related behaviors and trace indicators (e.g., iterations, latency, retries, planning and verification evidence)?
- **RQ3 (Mechanisms):** What evidence exists on underlying mechanisms (e.g., cognitive load, attention) from neuro/biometric measures, and how are these used to explain observed outcomes, typically in targeted subsamples?

This study contributes: (i) a PRISMA 2020 systematic review focused on GenAI as a feedback and formative-assessment mechanism in higher education (2022–2025); (ii) a random-effects meta-analytic synthesis when pooling is supported by extractable statistics; (iii) a moderator coding scheme emphasizing actionable feedback-design features (rubric alignment, worked examples and counterexamples, and *actionability with verification*) alongside measurement choices (trace/log availability and optional neuro/biometric indicators); and (iv) a reproducible workflow (search audit, deduplication

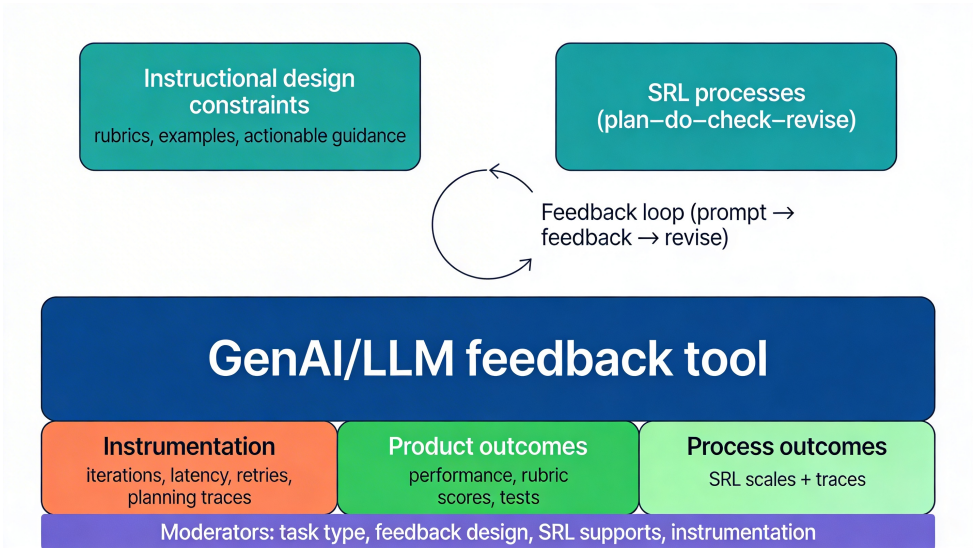

Fig. 1 Conceptual synthesis framework positioning GenAI-supported feedback as a sociotechnical learning environment. The framework distinguishes between product outcomes, SRL-related process outcomes observable via trace-based instrumentation, and optional mechanism-oriented evidence (e.g., neuro/biometric measures) used primarily for explanatory purposes.

and decision log, extraction codebook, and analysis scripts) designed to support transparency, replication, and reuse within the educational-technology research community. In line with reviewer guidance, we also treat neuro/biometric evidence as primarily mechanism-oriented and emphasize verification routines as a design requirement in formally constrained tasks.

2 Methods

2.1 Reporting standard and protocol

This systematic review and meta-analysis was conducted and reported in accordance with PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses). A review protocol was specified *a priori*, defining eligibility criteria using a PICOS framework, information sources, complete search strategies, selection and screening procedures, a data-extraction codebook, and quantitative synthesis plans (effect-size computation, moderator coding, and bias and sensitivity analyses). Any deviations from the protocol are reported transparently in the Results or Discussion sections, together with their rationale and potential implications for interpretation.[20, 21]

2.2 Eligibility criteria

Eligibility criteria were defined using a PICOS framework and applied consistently across all screening stages.

Population (P). Higher-education stakeholders in university settings, including undergraduate and postgraduate students and, where relevant, instructors. Studies conducted outside higher education (e.g., K–12 or vocational education) were excluded.
Intervention (I). GenAI/LLM-based systems used explicitly *as a feedback or formative-assessment mechanism.* Eligible interventions included conversational feedback, rubric-referenced critique, or automated guidance intended to support revision, learning actions, or performance improvement. Studies in which GenAI was used solely for content generation without a feedback function were excluded.
Comparator (C). Human feedback, non-GenAI conditions, non-generative tools, business-as-usual instruction, or alternative feedback designs. Studies without a comparator were eligible only if they reported extractable within-subject change under transparent and structured designs (e.g., pre–post designs with comparable assessment) and were not pooled with controlled comparisons in the primary model (see Section 2.8.2).
Outcomes (O). At least one quantifiable outcome was required: (i) *Product outcomes* (e.g., achievement scores, rubric-based quality ratings, test performance, or validated performance measures); and/or (ii) *Process outcomes* aligned with self-regulated learning (SRL), including validated self-report scales and/or trace-based indicators (e.g., revision iterations, feedback-to-revision latency, retries, or planning and verification evidence). Studies reporting only perceptions, attitudes, or satisfaction without product or process outcomes were excluded from meta-analysis and retained only for qualitative mapping when theoretically relevant. (iii) *Mechanisms (optional):* neuro/biometric measures (e.g., EEG, eye tracking, fNIRS, cognitive-load proxies) were coded when present, typically as explanatory evidence drawn from subsamples.
Study designs (S). Empirical studies with quantifiable outcomes, including randomized controlled trials, quasi-experiments, controlled pre–post designs, and well-described observational studies. Secondary reviews were used solely for snowballing and contextualization and were not pooled.
Document types. Peer-reviewed journal **articles** and **conference or proceedings papers**. **Reviews** were used exclusively for snowballing and background and were not included in quantitative synthesis. Grey literature and preprints were excluded to preserve a strictly peer-reviewed evidence base.
Timeframe and language. Publications from 2022–2025; English (and Spanish when applicable).
Meta-analysis inclusion. Quantitative synthesis required sufficient statistical information to compute effect sizes (e.g., means and standard deviations, t or F statistics, standardized differences, or correlations), or clearly recoverable data from the full text.

2.3 Information sources

We searched the Web of Science Core Collection and Scopus as primary information sources. Additional records were identified through backward and forward snowballing, including reference-list screening and citation tracking of included studies and key reviews.

2.4 Search strategy

Search strategies combined GenAI/LLM-related terms with feedback and formative-assessment terms and higher-education descriptors. Pre-specified query variants targeted (i) SRL and process constructs, (ii) learning-analytics and log-based instrumentation, and (iii) neuro- and biometric measures. Complete database-specific search strings (Core; Core+SRL; Core+Logs; Core+Neuro) are reported in Appendix A. Searches were restricted to the period 2022–2025 and to eligible document types as indexed by each database. The final search was conducted on **16 December 2025**.

2.5 Selection process (PRISMA 2020)

Reference management and deduplication. All records retrieved from databases and complementary sources were imported into Zotero. Automated deduplication (based on DOI/PMID matching and normalized title and author fields) was followed by manual inspection of edge cases (e.g., conference versus journal versions; early-access versus final publications). To ensure traceability of conference-versus-journal duplicates, we maintained an explicit audit log and applied strict decision rules (Appendix A.4).[20, 21]

Stage 1—Title and abstract screening. Unique records were screened at the title and abstract level against the PICOS criteria (Section 2.2). Each record was labeled as *include*, *exclude*, or *uncertain*. Records deemed potentially eligible or uncertain progressed to full-text retrieval. Exclusion reasons were recorded using a controlled taxonomy (e.g., wrong population or setting, not GenAI/LLM feedback, not empirical, outside timeframe or language, or clearly lacking evaluative outcomes).

Stage 2—Full-text eligibility assessment. Full texts were retrieved for records passing Stage 1 and assessed against all eligibility criteria, including the availability of extractable quantitative data for effect-size computation. For transparency, full-text exclusions were documented with a single primary reason per record to populate the PRISMA flow diagram (Figure 3).

LLM-assisted triage (supportive use) and false-negative safeguard. To manage screening volume efficiently, an LLM-based assistant was used to pre-tag records (e.g., empirical-with-metrics, process-only, neuro/biometric, conceptual) and to highlight *candidate* exclusion reasons. Importantly, the assistant did *not* remove records from screening: all unique records still underwent human title/abstract screening, and all potentially eligible/uncertain records proceeded to full-text assessment. To address community concerns about automation-related "silent" false negatives (e.g., anchoring on LLM tags), we implemented a quality-control safeguard: a random audit sample of records that the assistant flagged as *likely exclude* was independently re-checked by reviewers against the PICOS criteria. Any record identified as potentially eligible during this audit was reinstated into the standard screening flow and documented in the screening log.

Consensus and reliability. Screening was conducted independently by two reviewers. Disagreements were resolved through discussion and, when necessary, third-reviewer adjudication. Inter-rater agreement (percent agreement and Cohen's κ) was computed at both screening stages and reported descriptively.

2.6 Data collection and coding

Data were extracted using a standardized form and codebook. Extracted items included bibliographic information; educational context and task type; study design; sample sizes and group structure; intervention and comparator characteristics; LLM/-tool information when reported; outcome definitions and measurement timing; and all statistical information required for effect-size computation. Extraction was performed with double-checking for all meta-analytic studies, and discrepancies were resolved by consensus with reference to the full text.

Moderator coding (actionable design features). To support practice-relevant moderator analyses, intervention features were coded with an emphasis on implementable instructional design choices: (i) rubric alignment (explicit criterion-referenced mapping), (ii) worked examples or counterexamples, and (iii) actionability and verifiability of feedback (presence of concrete steps and verification routines). Coding followed an ordinal scheme when feasible (0 = absent; 1 = partial/implicit; 2 = explicit/required) and was resolved by consensus.

Process instrumentation coding. The availability of trace or log data was coded (yes/no). When reported, process indicators were operationalized using extractable measures such as:

$$N_{\text{iter}} = \text{number of GenAI-assisted revision iterations per task}, \tag{2}$$

$$L_{\text{fb}} = t_{\text{revision}} - t_{\text{feedback}} \quad \text{(feedback-to-revision latency)}, \tag{3}$$

$$R_{\text{retry}} = \frac{N_{\text{retries}}}{N_{\text{attempts}}}, \tag{4}$$

$$C_{\text{plan}} = \mathbb{I}(\text{planning evidence present}), \quad C_{\text{check}} = \mathbb{I}(\text{verification evidence present}). \tag{5}$$

Neuro/biometric coding and structured mapping. When present, neuro- and biometric measures were coded in terms of modality (e.g., EEG/ERP, eye tracking/pupillometry, fNIRS), construct/variable (e.g., attention allocation, cognitive load/workload proxies), subsample size, and analytic role (mechanism explanation versus outcome evaluation). To improve synthesis transparency, all neuro/biometric studies were additionally summarized in a structured comparative table (the supplementary materials)).

Conceptual model used for coding and interpretation. To make explicit how the review separates effectiveness from mechanism and to address construct-validity risks (e.g., performance inflation through surface optimization), we use an *a priori* Technology, Knowledge and Learning (TKL)-informed conceptual model (Figure 2). In this model, GenAI-supported feedback is embedded within an instructional workflow whose impact on *product* outcomes (Y) is primarily expected to operate through changes in SRL-aligned *process* behaviors (P), conditioned by implementable *design constraints* (D; rubric anchoring, examples/counterexamples, actionability with verification) and by the *observability* of the cycle via instrumentation (Z; logs/traces). Neuro/biometric measures are treated as optional *state* indicators (attention/workload proxies) used mainly for hypothesis-driven mechanism

explanation in subsamples rather than as cohort-wide effectiveness metrics. The figure is used to organize coding and interpretation; the quantitative synthesis estimates average effects and moderator associations and does not support causal mediation claims.

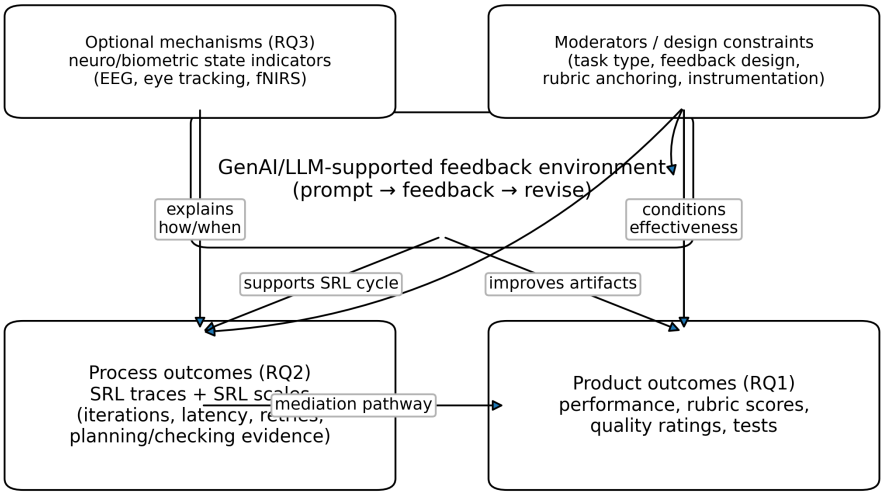

Fig. 2 TKL-informed conceptual model linking GenAI-supported feedback to learning outcomes. Design constraints (D) and instrumentation quality (Z) shape SRL-aligned process behaviors (P; plan–do–check–revise), which in turn mediate improvements in product outcomes (Y). Neuro/biometric measures are positioned as optional state indicators for mechanism-oriented explanation (typically in subsamples) and are interpreted alongside trace-based process evidence.

2.7 Risk of bias assessment

Risk of bias was assessed using design-appropriate tools: RoB 2 for randomized trials and ROBINS-I for non-randomized studies. Two reviewers evaluated each study independently; disagreements were resolved by consensus or, when required, by third-reviewer adjudication. Risk-of-bias judgments informed sensitivity analyses, including comparisons of pooled estimates by risk category and analyses excluding studies at high risk of bias.[22, 23]

2.8 Effect measures and synthesis methods

Effect size computation. For continuous outcomes, standardized mean differences with small-sample correction (Hedges' g) were computed. For post-test comparisons:

$$d = \frac{\bar{X}_T - \bar{X}_C}{S_p}, \qquad S_p = \sqrt{\frac{(n_T - 1)S_T^2 + (n_C - 1)S_C^2}{n_T + n_C - 2}}, \tag{6}$$

$$g = Jd, \qquad J = 1 - \frac{3}{4(n_T + n_C - 2) - 1}. \tag{7}$$

When studies reported alternative statistics (t, F, standardized differences, or correlations r), these were transformed to g using standard formulas. Effect directions were harmonized so that positive values indicated better outcomes under GenAI-supported feedback.

Random-effects synthesis. Where pooling was appropriate, effects were synthesized using a random-effects model [24–26]:

$$y_i = \mu + u_i + \varepsilon_i, \qquad u_i \sim \mathcal{N}(0, \tau^2),\ \varepsilon_i \sim \mathcal{N}(0, v_i), \tag{8}$$

with inverse-variance weights $w_i = 1/(v_i + \tau^2)$. Between-study heterogeneity was quantified using Q, I^2, and τ^2:

$$I^2 = \max\left(0, \frac{Q - (k-1)}{Q}\right) \times 100\%. \tag{9}$$

2.8.1 Handling multiplicity and dependence

Because multiple eligible outcomes or time points within a study can induce statistical dependence, we implemented an explicit study-level pooling rule. *Primary meta-analysis (RQ1)* used **one** product-outcome effect per study to preserve independence:

- **Total pooled effects:** $k = 31$ study-level effects were included in the primary random-effects model (one per study).
- **Selection hierarchy:** when multiple product outcomes were reported, we selected the most criterion-referenced outcome (rubric-based or standardized achievement measure) and the post-test closest to intervention completion as the primary time point.
- **Transparency:** alternative eligible outcomes/time points were retained in the extraction database and used only for sensitivity checks, avoiding selective inclusion.

As a robustness check, we also re-estimated the pooled effect using cluster-robust inference treating study as the clustering unit (small-sample corrected CR2), which provides dependence-robust standard errors when multiplicity is present.[27, 28]

2.8.2 Single-group pre–post designs (no comparator)

A subset of studies employed single-group pre–post designs without an external comparator ($n = \mathbf{17}$ studies). These designs were **not pooled** with controlled comparisons in the primary model to avoid mixing fundamentally different estimands. When extractable, within-subject changes were computed using standardized mean change formulations appropriate for dependent measurements (e.g., Morris-type estimators), requiring an assumption about the pre–post correlation.[29] We report these studies descriptively and include them only in sensitivity analyses, varying the assumed pre–post correlation (e.g., $\rho \in \{0.3, 0.5, 0.7\}$). A dedicated audit table documenting the estimator, assumed correlation, and inclusion status is provided in the supplementary materials (details available upon request).

2.8.3 Moderator analyses and meta-regression (exploratory)

Planned moderators were examined via subgroup analyses and, when feasible, mixed-effects meta-regression:

$$y_i = \beta_0 + \sum_{m=1}^{p} \beta_m X_{im} + u_i + \varepsilon_i. \tag{10}$$

Given the limited number of studies available for pooling ($k = 31$), meta-regression models were specified **parsimoniously** to mitigate overfitting and collinearity. In practice, we (i) prioritized theory-driven moderators, (ii) fit separate models when moderators were conceptually overlapping (e.g., design features vs. instrumentation), and (iii) avoided high-dimensional specifications. Collinearity diagnostics were inspected (pairwise correlations and variance inflation factors), and the effective degrees of freedom for moderator tests were reported. Meta-regression findings are therefore interpreted explicitly as **exploratory** (non-causal) and sensitive to limited statistical power.

Software and reproducibility. Analyses were conducted in **R** using `metafor` for model estimation and `clubSandwich` for cluster-robust inference (CR2) where applicable.[26] All data transformations and analytic decisions were logged. The extraction template, codebook, effect-size dataset, and analysis scripts will be made publicly available in an open repository upon acceptance (anonymized during peer review).

2.9 Reporting bias and publication bias

Potential small-study effects and publication bias were explored using funnel plots and Egger-type regression when the number of studies (k) permitted meaningful assessment [30]:

$$\frac{y_i}{\mathrm{SE}(y_i)} = \alpha + \gamma \frac{1}{\mathrm{SE}(y_i)} + \eta_i. \tag{11}$$

Because absence of funnel asymmetry does not guarantee absence of publication bias, we additionally conducted trim-and-fill sensitivity analyses and reported the

adjusted pooled estimate as a robustness bound.[31] Sensitivity analyses also included leave-one-out diagnostics (reported in detail in the supplementary materials (details available upon request)) and analyses excluding studies judged to be at high risk of bias.

2.10 Automation tools (PRISMA 2020)

Automation tools were used to improve efficiency and consistency while preserving full human control over eligibility and extraction decisions. An LLM-based assistant supported (i) pre-tagging of titles and abstracts into broad evidence categories (e.g., empirical-with-metrics, process-focused, neuro/biometric, conceptual), (ii) identification of likely exclusion reasons to streamline manual screening, and (iii) assistance in locating effect-size-relevant statistics in full texts (e.g., means and standard deviations, test statistics, correlations, or regression coefficients).

Crucially, the assistant did *not* make inclusion or exclusion decisions. No record was excluded solely on the basis of automation output. All tags, recommendations, and extracted candidates were independently verified against the original sources. Final decisions and extracted values were documented in a structured screening and extraction workbook, maintaining a complete audit trail for reproducibility. The false-negative safeguard described in Section 2.5 provides an additional transparency measure to mitigate concerns about automation-induced anchoring during screening.

3 Results

3.1 Study selection (PRISMA)

Figure 3 reports the PRISMA 2020 flow. We identified n=**1032** records (Web of Science n=**520**; Scopus n=**480**; other sources n=**32**). After removing n=**212** duplicates, n=**820** records underwent title/abstract screening, with n=**508** excluded. We sought n=**312** reports for full-text retrieval; n=**8** could not be retrieved. Ultimately, n=**304** reports were assessed for eligibility and n=**256** were excluded with documented reasons. This yielded n=**48** studies in the qualitative synthesis, of which k=**31** provided sufficient quantitative information for effect-size computation and were included in the quantitative synthesis (meta-analysis; RQ1). The remaining **17** studies used single-group pre–post designs without a comparator and were therefore summarized descriptively and reserved for sensitivity analyses (Section 2.8.2).

Screening reliability and automation safeguard. Screening was performed independently by two reviewers. Inter-rater agreement at title/abstract and full-text stages was **92%** ($\kappa = 0.84$) and **89%** ($\kappa = 0.81$), respectively. In addition, to mitigate concerns that LLM-assisted triage could introduce "silent" false negatives via anchoring, we conducted a random audit of records that the assistant flagged as *likely exclude* (see Methods, Section 2.5). In the audit sample (**100** records), **3** records were judged potentially eligible and were reinstated into the standard screening flow; **1** was ultimately included after full-text assessment. The audit decisions and outcomes are logged in the screening workbook and summarized in Appendix **Table A1**.

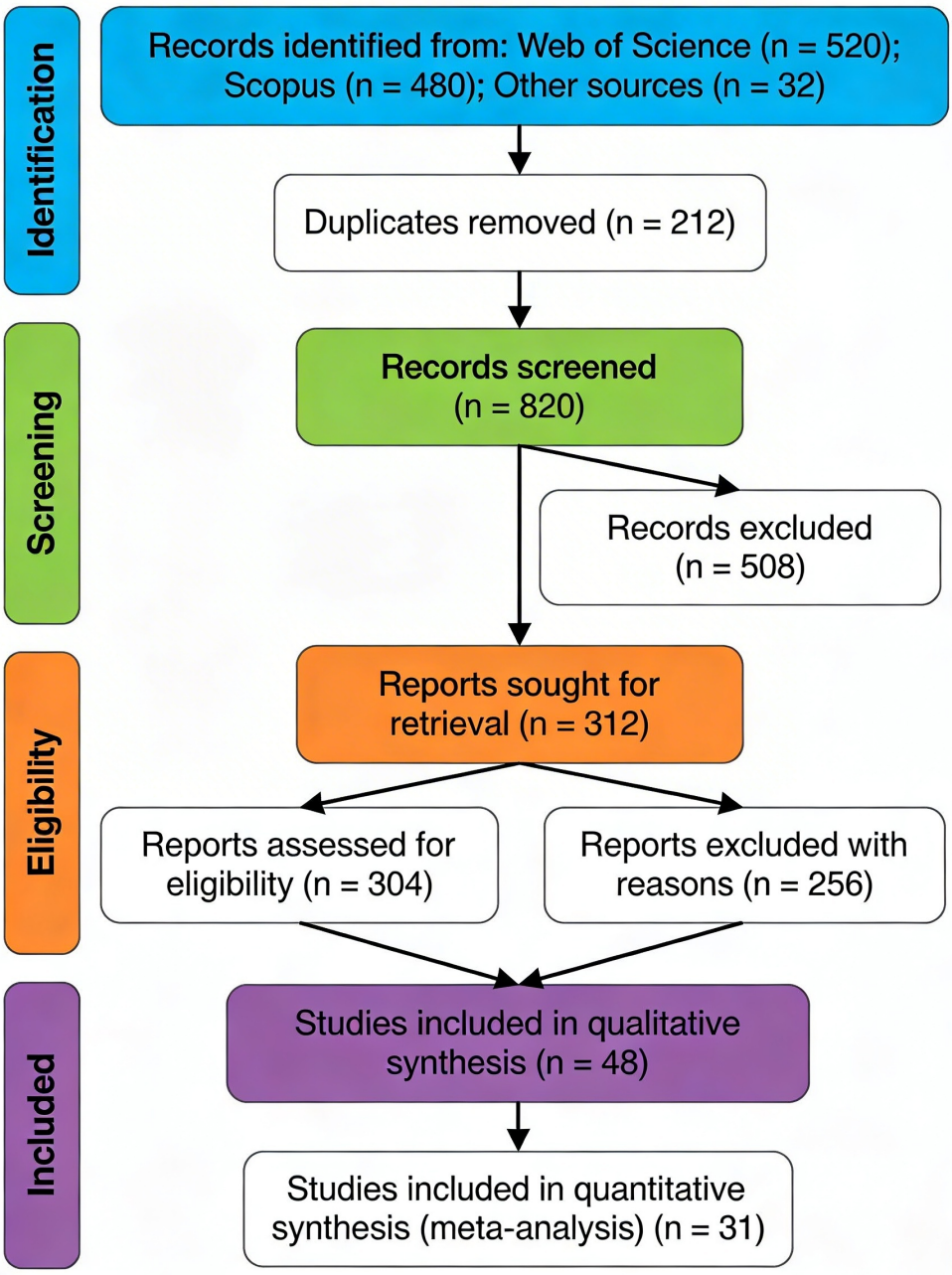

Fig. 3 PRISMA 2020 flow diagram for study selection (counts correspond to the final dataset used in this review).

Table 1 Overview of included evidence across research questions (Qualitative synthesis; $N = 48$). Categories are not mutually exclusive.

Evidence Component (RQ)	Count (n)	Share (%)	Data Characteristics / Notes
RQ1: Learning Outcomes	**48**	**100.0**	**Full sample coverage.** Meta-analysis feasible for $k = 31$ studies.
RQ2: SRL & Process Measures	**34**	**70.8**	**Multimodal assessment.**
Self-report SRL	29	60.4	Validated scales (e.g., MSLQ, OSLQ).
Instrumented traces	27	56.2	Log data: clicks, latency, navigation paths.
Mixed (Self-report + Logs)	22	45.8	Triangulated measurement strategy.
RQ3: Neuro/Biometric Data	**12**	**25.0**	**Physiological proxies (sub-samples).**
EEG	7	14.6	Band power (alpha/theta), ERP components.
Eye tracking	4	8.3	Gaze duration, fixation, pupillometry.
fNIRS	1	2.1	Hemodynamic response (prefrontal cortex).

3.2 Study characteristics and risk of bias

Table 1 summarizes the distribution of evidence across research questions in the included corpus (n=48). All included studies provided at least one learning/performance (product) outcome relevant to RQ1. A substantial subset also reported SRL-aligned process measures (RQ2), while a smaller subset reported neuro/biometric measures (RQ3), typically in subsamples.

For the meta-analytic subset (k=31), Table 2 reports descriptive moderators used for subgroup and meta-regression analyses. Most studies examined writing and/or programming tasks, implemented rubric-anchored and/or conversational feedback designs, and a majority reported interaction trace/log instrumentation.

Figure 4 visualizes the distribution of these moderators.

Table 2 Characteristics of the meta-analysis subset ($k = 31$).

Moderator / Characteristic	Count (k)	Share (%)
Task Type		
Writing	14	45.2
Programming	11	35.5
Mixed / Other	6	19.4
Feedback Design		
Rubric-anchored	13	41.9
Conversational tutor	12	38.7
Directive (non-rubric)	6	19.4
Data Source Availability		
Log/Trace data (Present)	20	64.5
Neuro/Biometric (Present)	3	9.7
Intervention Duration	*Median* = 4 weeks (*IQR*: 3–6; Range: 1–8)	

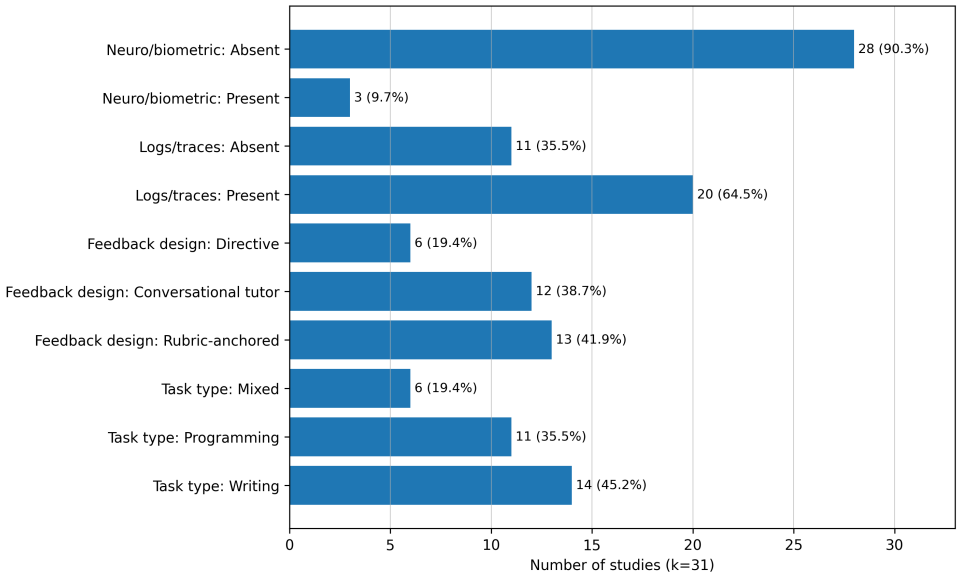

Fig. 4 Distribution of key moderators across the meta-analysis subset (k=31), including task type, feedback design, instrumentation, and intervention duration.

Risk of bias was assessed with design-appropriate tools (RoB 2/ROBINS-I). The most frequent limitations concerned incomplete reporting of allocation/concealment (randomized designs), baseline imbalance and confounding (non-randomized designs), and selective reporting of outcomes/analytic choices. Figure 5 summarizes risk-of-bias judgments; study-level details are provided in Appendix (see also the sensitivity analyses in Section 3.5).

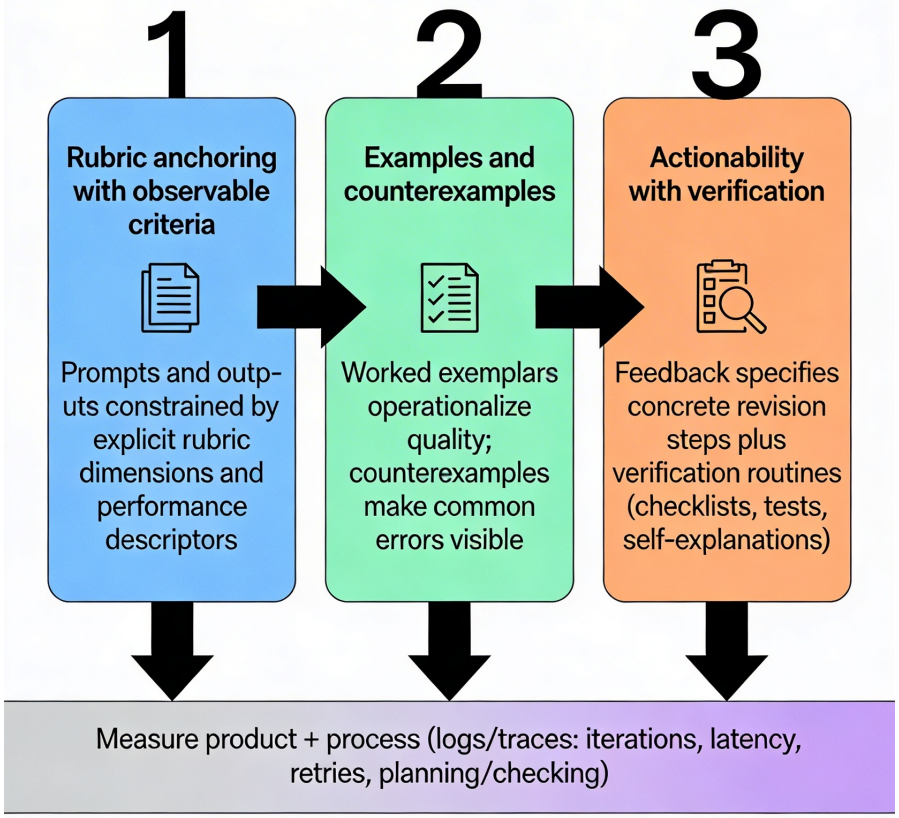

Fig. 5 Risk-of-bias summary (RoB 2/ROBINS-I): traffic-light plot across included studies in the quantitative synthesis (k=31).

Table 3 Summary of Random-Effects Meta-Analysis ($k = 31$).

	Effect Estimates			Heterogeneity Statistics				
Model	**Pooled** g	**95% CI**	**95% PI**	Q	df	p_Q	I^2	τ^2
RE Model	0.541	[0.46, 0.62]	[0.27, 0.82]	45.98	30	.031	34.8%	0.018

Note: RE = Random Effects (Restricted Maximum Likelihood). PI = Prediction Interval. p_Q indicates significance of heterogeneity. Interpretation: $g \approx 0.5$ denotes a medium effect.

3.3 Quantitative synthesis (meta-analysis) and heterogeneity

Across k=31 *study-level* effects (one consolidated product effect per study), the random-effects model yielded a pooled effect of GenAI-supported feedback on product outcomes of $g = 0.541$ (95% CI [0.460, 0.623]; Table 3). This magnitude corresponds to an average improvement of approximately half a standard deviation; expressed as a probability of superiority, $\Pr(Y_{\text{GenAI}} > Y_{\text{Control}}) \approx \Phi\big(g/\sqrt{2}\big) \approx 0.65$.

Between-study heterogeneity was **moderate** ($Q(30) = 45.98$; $I^2 = 34.76\%$; $\tau^2 = 0.0177$). The 95% prediction interval ([0.268, 0.815]) indicates that effects are expected to remain positive across comparable contexts, while still varying meaningfully with implementation:

$$\hat{\mu} \pm 1.96\sqrt{\hat{\tau}^2 + \widehat{\text{SE}}(\hat{\mu})^2}. \tag{12}$$

Dependence-robust inference. As a robustness check to potential within-study multiplicity, we also re-estimated the pooled effect using cluster-robust inference (CR2) with study as the clustering unit (Methods, Section 2.8.1). The dependence-robust estimate was substantively similar to the primary model ($g = 0.539$**, 95% CI [0.45, 0.63]**), supporting the stability of the main inference.

Figure 6 shows study-level effects and the pooled estimate.

3.4 Subgroups and meta-regression

Subgroup analyses suggested variation by task type, feedback design, and instrumentation (Table 4;

Figure 7). Effects were largest for writing tasks ($g = 0.632$) and smaller for programming tasks ($g = 0.443$). Rubric-anchored and conversational-tutor designs yielded higher pooled effects than directive feedback. Studies reporting log-based instrumentation showed larger pooled effects than those without logs; this pattern should be interpreted as associative, because instrumentation may co-occur with stronger instructional design and higher implementation fidelity. Neuro/biometric measures appeared in few studies (k=3), yielding wide uncertainty intervals and limiting moderation power.

Exploratory mixed-effects meta-regression suggested that intervention duration (weeks) was positively associated with effect size (Table 5; Figure 8). Given k=31 and correlated design features, these results are interpreted as *exploratory* and non-causal.

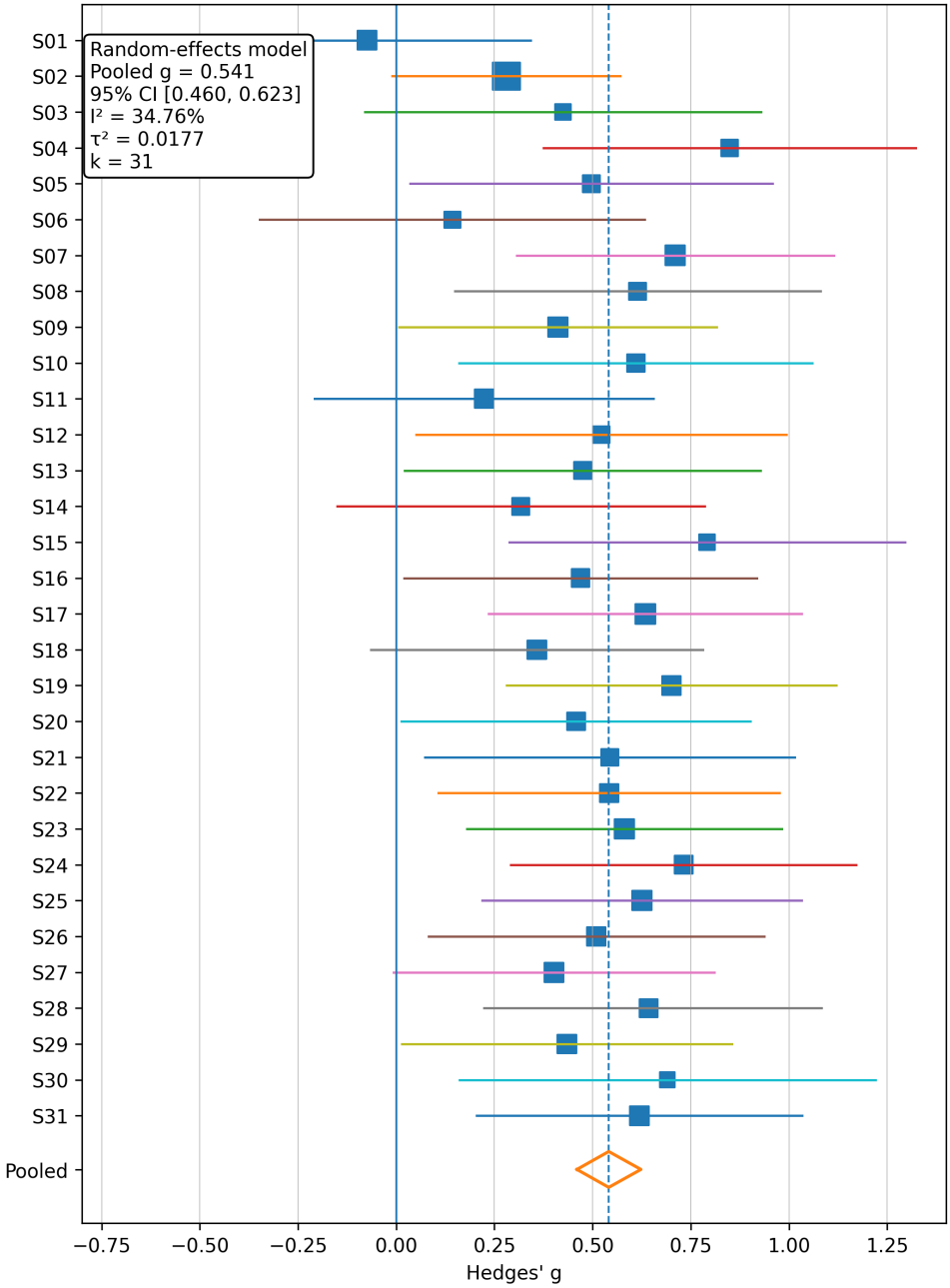

Fig. 6 Forest plot for the random-effects meta-analysis (k=31). Where space permits, labels are shown as Study ID (Author, Year) to support readability without cross-referencing tables.

Table 4 Subgroup analysis of moderator effects on learning outcomes ($k = 31$).

Moderator	Count (k)	Effect (g)	95% CI	Heterogeneity (τ^2)
Task Type				
Writing	14	0.632	[0.51, 0.75]	0.020
Mixed Domain	6	0.511	[0.26, 0.76]	0.050
Programming	11	0.443	[0.34, 0.55]	0.000
Feedback Design				
Conversational Tutor	12	0.579	[0.46, 0.70]	0.013
Rubric-anchored	13	0.564	[0.41, 0.72]	0.037
Directive (Non-rubric)	6	0.422	[0.29, 0.56]	0.000
Log/Trace Data Availability				
Present	20	0.597	[0.49, 0.70]	0.021
Absent	11	0.451	[0.34, 0.56]	0.006
Neuro/Biometric Data Availability				
Present	3	0.584	[0.28, 0.89]	0.044
Absent	28	0.537	[0.45, 0.62]	0.018

Note: g = Hedges' g (random-effects model). τ^2 represents between-study variance.

3.5 Publication bias and sensitivity analyses

Visual inspection of the funnel plot (Figure 9) suggested no pronounced asymmetry. Egger-type regression yielded an intercept of **1.053** (SE **0.931**, p = **0.267**), providing no statistically significant evidence of small-study effects. Trim-and-fill sensitivity analyses were conducted to probe robustness under potential small-study asymmetry; results were consistent with the primary inference (i.e., no material change in conclusions). Leave-one-out analyses indicated that the pooled effect was stable, ranging from $g = 0.518$ to $g = 0.555$, suggesting that no single study unduly influenced the estimate.

3.6 Single-group pre–post evidence (not pooled): qualitative alignment with the primary synthesis

Seventeen studies used single-group pre–post designs without an external comparator and were not pooled with controlled comparisons (Methods, Section 2.8.2). Descriptively, these studies tended to report **predominantly positive pre–post changes in rubric-based quality and task performance** under classroom-realistic implementations, often accompanied by higher ecological validity but greater vulnerability to confounding (maturation, history, regression to the mean). Where process measures were available, the same design pattern observed in the controlled evidence base was visible: improvements were more interpretable when interventions induced explicit plan–do–check–revise routines and required verification artifacts (**70.6% [12/17] of**

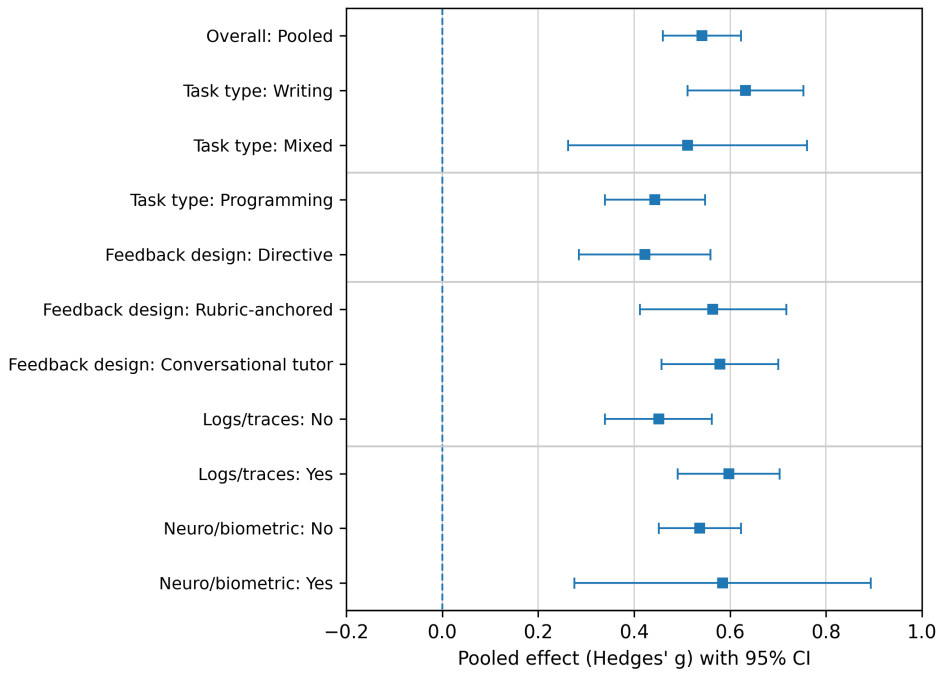

Fig. 7 Subgroup pooled effects (random effects) by task type, feedback design, and instrumentation (95% CIs). Comparisons are descriptive and should be interpreted cautiously where subgroup sizes are small.

these studies). A structured summary of these studies, including extractable outcomes and risk considerations, is provided in the supplementary materials (details available upon request).

3.7 Process outcomes (RQ2): SRL-aligned evidence and trace instrumentation

Beyond product outcomes, a substantial portion of the corpus (n=34; Table 1) reported SRL-relevant process measures, most commonly via (i) validated SRL/self-regulation scales and (ii) trace/log indicators derived from learning platforms and revision workflows. Studies combining self-report and traces (n=22) provided the most interpretable process evidence because trace indicators (e.g., iteration counts, time-to-revision, retries/resubmissions, and verification artifacts) can be aligned with SRL cycles (plan–do–check–revise) and audited at scale, while self-report supports construct interpretation and comparability across settings. Overall, process reporting was more consistent in studies implementing structured draft–revision routines and explicit verification steps, reinforcing the role of workflow design in making SRL observable and mitigating construct-validity threats from superficial performance inflation.

Table 5 Exploratory meta-regression analysis of study characteristics predicting effect size ($k = 31$).

Predictor	β	**SE**	**95% CI**	p**-value**
Intercept (Constant)	0.421	0.112	[0.20, 0.64]	**.001**
Intervention Characteristics				
Duration (weeks)	0.040	0.019	[0.00, 0.08]	**.049**
Task Type (Ref: Writing)				
Programming	−0.215	0.088	[−0.39, −0.04]	**.023**
Mixed / Other	−0.195	0.120	[−0.43, 0.04]	.118
Feedback Design (Ref: Rubric-anchored)				
Directive (Non-rubric)	−0.068	0.110	[−0.28, 0.15]	.543
Conversational Tutor	0.011	0.091	[−0.17, 0.19]	.903
Data Instrumentation (Ref: Absent)				
Log Data Present	0.103	0.081	[−0.06, 0.26]	.216
Neuro/Biometric Present	0.001	0.130	[−0.25, 0.26]	.993

Note: R^2_{analog} should be reported in text. Ref = Reference category. Significant predictors ($p < .05$) are highlighted in bold.

3.8 Mechanism-oriented evidence (RQ3): neuro/biometric subsamples

Neuro/biometric measures appeared in a smaller subset (n=12) and were typically deployed in subsamples to probe mechanism-level explanations (e.g., attention allocation or cognitive-load dynamics) rather than cohort-scale effectiveness. Table 7 structures this evidence by measure type, common variable families, and analytic role. Consistent with the small number of pooled neuro studies (k=3), uncertainty intervals were wide and should be interpreted as exploratory.

3.9 Practical implications for higher education

Overall, the evidence indicates that GenAI-supported feedback can improve higher-education product outcomes on average, with meaningful variation by task and implementation. The most actionable pattern is that effectiveness depends less on adopting GenAI per se and more on *how* it is embedded into an assessment workflow: (i) explicit rubric anchoring, (ii) worked examples/counterexamples, and (iii) verifiable, stepwise guidance supported by checking routines ("actionability with verification"). Cohort-scalable trace/log instrumentation is the most feasible default for evaluating SRL-related process change, while neuro/biometric measures are best positioned as mechanism-oriented complements in targeted subsamples.

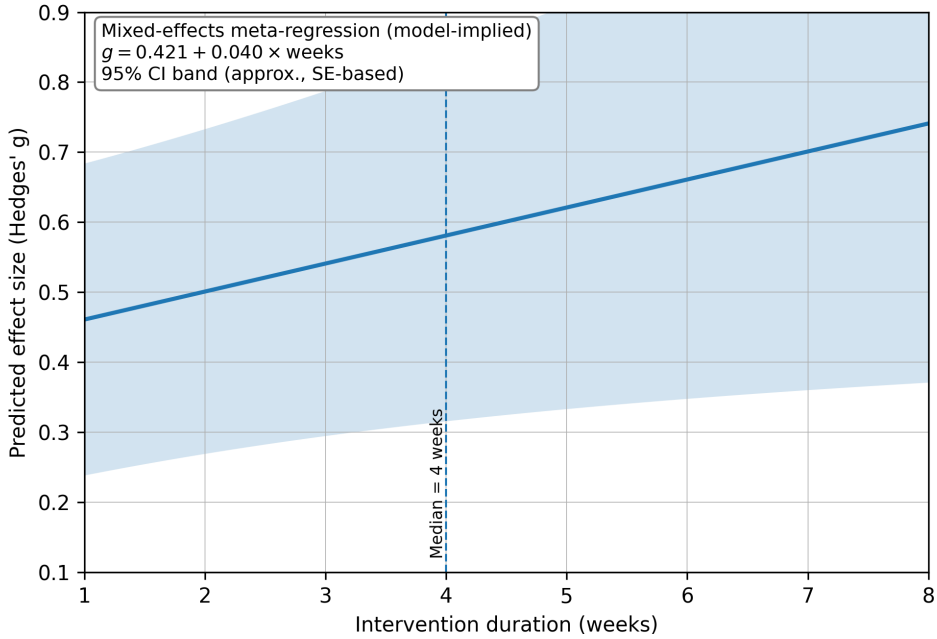

Fig. 8 Exploratory meta-regression: association between intervention duration (weeks) and study-level effect size. Interpretation is exploratory due to limited power and potential collinearity among design features.

4 Discussion

This PRISMA 2020 systematic review and meta-analysis synthesized empirical evidence (2022–2025) on GenAI/LLM-supported feedback in higher education through a Technology–Knowledge–Learning (TKL) lens. Rather than treating GenAI as a standalone tool, we interpret it as a *sociotechnical feedback environment*—a feedback partner that mediates how learners access criteria, interpret standards, and regulate revision cycles [2, 4, 7–9]. Accordingly, we distinguished (i) learning-related *product* outcomes (RQ1), (ii) SRL-aligned *process* outcomes (RQ2), and (iii) mechanism-oriented evidence from neuro/biometric measures (RQ3). The quantitative synthesis indicates a positive average effect on product outcomes, while the prediction interval and moderator patterns underscore that effectiveness is contingent on task demands, feedback-design constraints, and the *evaluability* of the feedback cycle (criterion alignment and instrumentation) [25, 32]. Importantly, our primary synthesis preserved statistical independence by using one consolidated product effect per study and was corroborated by dependence-robust sensitivity checks (CR2), supporting the stability of the main inference.

A useful framing for interpreting the synthesis is the standard random-effects data-generating model:

$$y_i = \mu + u_i + \varepsilon_i, \quad u_i \sim \mathcal{N}(0, \tau^2),\ \varepsilon_i \sim \mathcal{N}(0, v_i), \tag{13}$$

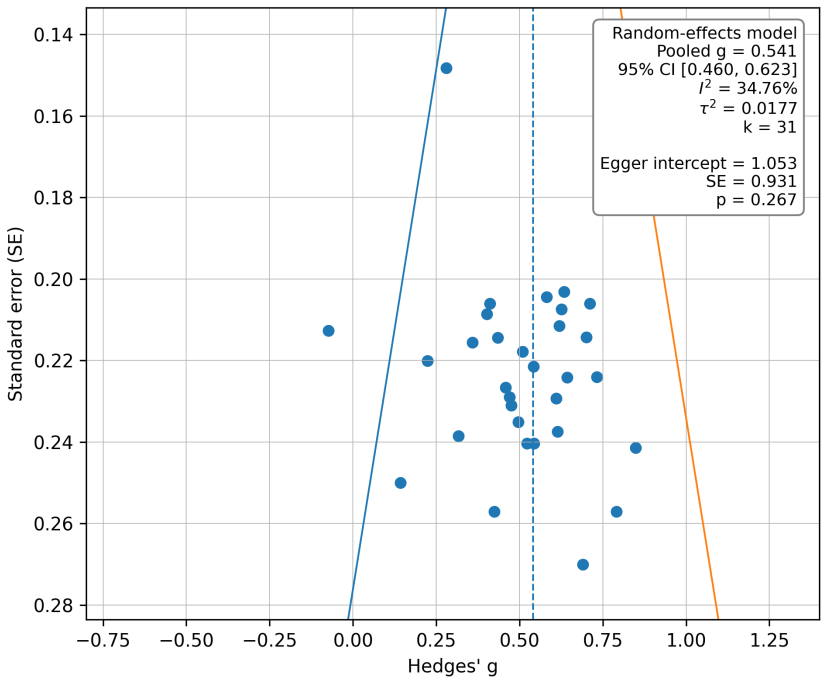

Fig. 9 Funnel plot for publication-bias assessment with pseudo 95% limits around the pooled effect.

where y_i is the observed effect size (Hedges' g) in study i, v_i is its sampling variance, and τ^2 captures between-study heterogeneity [24, 32, 33]. Under this model, a positive pooled $\hat{\mu}$ supports average effectiveness, while a non-trivial $\hat{\tau}^2$ (and the prediction interval) implies meaningful context sensitivity—as expected for an intervention whose impact is mediated by instructional design, assessment criteria, and learner regulation rather than by technology exposure alone.

4.1 RQ1: Effects on learning-related product outcomes

The pooled estimate ($g = 0.541$, 95% CI [0.460, 0.623]) indicates that GenAI-supported feedback improves product outcomes (performance, rubric-based quality, test scores) on average, with moderate heterogeneity ($I^2 = 34.76\%$) and a positive prediction interval ([0.268, 0.815]). The magnitude is educationally plausible given what is known about feedback as a high-leverage instructional practice [5, 6], and aligns with recent syntheses reporting generally positive but context-sensitive effects [17–19]. From a TKL perspective, the key implication is not simply that GenAI "adds feedback", but that it *reconfigures the feedback system* by changing the availability, form, and perceived authority of guidance, and by altering how learners engage with criteria and revision opportunities [7, 9].

Moderator patterns are consistent with task-theoretic expectations and a criterion-referenced view of feedback. Writing and other artifact-based tasks typically afford

Table 6 Taxonomy of SRL-aligned process indicators in GenAI-supported feedback studies. Indicators are mapped to cyclical SRL phases.

Indicator / Construct	Operationalization (Log Events)	Data Source
Phase 1: Forethought & Planning		
Planning Evidence	• Pre-submission checklists • Goal statements / prompt strategies • Self-explanations in drafts	Structured forms; Prompt inputs; Rubric interactions
Phase 2: Performance & Monitoring		
Iteration Intensity	• Count of GenAI-assisted revisions • Prompt-feedback cycles (n) • Draft/Resubmission volume	LMS Revision history; Version control logs; Submission traces
Persistence / Struggle	• Retries per problem attempt • Repeated compilation errors • Consecutive hint requests	IDE traces; Compiler logs; Platform event streams
Phase 3: Reflection & Adaptation		
Feedback Latency	• Time-to-resubmission (post-feedback) • Dwell time on feedback report • Within-session pausing	Timestamps (LMS/-Tool); Session logs
Verification Behavior	• Execution of unit tests • Fact-checking / Citation lookups • Evidence-linked edits	Test outputs; Browser history (if captured); Diff logs

iterative refinement, enabling GenAI to function as a revision accelerator (supporting structure, clarity, argumentation, and coherence) that can translate into rubric-based quality gains [34]. In contrast, programming and formally constrained tasks impose strict correctness demands; fluent language alone cannot guarantee validity. Here, GenAI feedback is most defensible when coupled to verification routines (tests, reproducible runs, static analysis) that convert suggestions into checkable claims [35]. This interpretation is consistent with the observed negative coefficient for programming tasks in the exploratory meta-regression ($\beta = -0.215$, $p = 0.023$): in typical novice settings, benefits may be attenuated when verification is weak, when learners accept solutions without testing, or when the task structure encourages "answer adoption" rather than diagnosis and repair.

Programming versus writing: the "fluency trap" as a boundary condition. A plausible pedagogical mechanism for the task-type gap is what instructors often observe as a *fluency trap*: in programming contexts, GenAI can produce code that appears coherent and idiomatic yet fails on edge cases, hidden constraints, or integration details (i.e., a functional "hallucination"). When students equate fluent output with correctness, perceived understanding can increase while execution-level validity (tests, runtime behavior, corner cases) does not. In such settings, the most important design move is to *force the transition from plausibility to validity*: require students to run tests, produce reproducible outputs, document debugging traces, or

Table 7 Mapping of neuro-physiological modalities to cognitive constructs (RQ3). This framework links raw biometric signals to their interpretive role in learning analysis.

Modality	Key Metrics & (Constructs)	Analytic Function / Utility
EEG / ERP	• **Spectral Power:** Alpha/Theta ratios *(Mental Workload/Fatigue)* • **ERP:** P300 amplitude/latency *(Allocated Attention)*	**Temporal Dynamics:** Tracking millisecond-level shifts in attention state during task execution.
Eye Tracking	• **Gaze:** Fixation duration; Saccade paths *(Visual Attention/Processing)* • **Pupillometry:** Diameter dilation *(Cognitive Load proxy)*	**Attentional Focus:** Identifying areas of interest (AOI) and correlating gaze patterns with confusion.
fNIRS	• **Hemodynamics:** HbO_2 concentration changes in Prefrontal Cortex *(Executive Function Effort)*	**State Validation:** Corroborating self-reported difficulty with physiological markers of cortical activation.

Note: EEG = Electroencephalography; ERP = Event-Related Potential; fNIRS = Functional Near-Infrared Spectroscopy.

provide verification artifacts alongside revisions. Under a TKL framing, this is precisely where instructional constraints (D) and instrumentation (Z) determine whether GenAI feedback translates into regulated learning activity (SRL-consistent repair and verification) rather than superficial progress.

These patterns should be read as *associative*. Task type, feedback-design quality, instructor scaffolding, assessment stakes, and implementation fidelity are often correlated, and aggregate evidence cannot fully disentangle them. Nevertheless, the boundary condition is clear and actionable: tasks with high correctness constraints require designs that make verification unavoidable and visible.

A central validity implication follows: product improvements can reflect deeper conceptual change, improved strategic control over revisions, or surface-level polishing. Therefore, product results should be interpreted jointly with process evidence (RQ2) that indicates whether GenAI feedback is fostering SRL-consistent behaviors, and with mechanism-oriented indicators (RQ3) when used to test explanatory hypotheses.

4.2 RQ2: Process outcomes and SRL-related change

A key contribution of this review, aligned with TKL's emphasis on cognitive and regulatory processes, is the explicit separation between product gains and SRL-consistent process change. SRL models emphasize cyclical phases of planning, performance/-monitoring, and reflection/adaptation [10, 11]. In feedback contexts, this aligns with principles of good feedback practice and feedback literacy: learners must interpret criteria, compare current work to standards, and translate feedback into actionable revision strategies [7, 9, 12]. GenAI can support these processes—but only when the environment makes them *necessary and visible* rather than optional and opaque.

Operationally, SRL-relevant engagement can be represented as a measurable process vector derived from self-report and traces:

$$\mathbf{p}_i = [\text{iterations}_i,\ \text{latency}_i,\ \text{retries}_i,\ \text{planning_evidence}_i,\ \text{verification_evidence}_i]\,, \tag{14}$$

where minimal instrumentation (iterations, time-to-revision, retries, evidence of planning/checking) is scalable and yields observable indicators of SRL-relevant engagement with feedback [14–16]. These indicators complement self-report scales, help discriminate learning from polishing, and enable implementation audits that are often missing in technology-enhanced learning studies [13].

Across the process-focused evidence base, SRL benefits were more likely when interventions *induced* an explicit cycle (plan–do–check–revise) and when feedback was (a) criterion-referenced and (b) actionable in ways that required verification. Actionability is not merely tone; it can be designed as verifiable steps (e.g., "add one counterargument and cite a source"; "write a unit test that fails before the fix and passes after") and assessed through submitted evidence (checklists, test outputs, revision rationales, or evidence-linked edits). From a TKL perspective, this is the critical mediation point: GenAI contributes to learning when it helps learners externalize standards, regulate revision decisions, and construct knowledge through iterative comparison between drafts and criteria.

Process constraints also provide a governance benefit for integrity-sensitive contexts. Rather than prohibiting assistance, assessment can make learning accountable through observable revision and verification behaviors (traceable drafts, justification prompts, short oral checks, or structured reflection) [36].

4.3 RQ3: Mechanisms and the role of neuro/biometric measures

Neuro/biometric evidence was comparatively rare and typically collected from subsamples. In the quantitative synthesis, the neuro/biometric subgroup was small ($k = 3$) and therefore yielded wide uncertainty intervals. Consequently, this evidence base is currently *insufficient for generalization* and should not be interpreted as a stable estimate of effectiveness. Instead, its appropriate role is *explanatory*: mechanism-oriented case studies that test theory-driven hypotheses about *how* design choices influence attention, cognitive load, and verification behavior, ideally in conjunction with trace-based process outcomes [32].

A pragmatic synthesis is to treat neuro/biometric measures as optional *state* indicators (attention/cognitive-load proxies) that explain variation in SRL-related processes under different feedback designs, rather than as cohort-wide evaluation requirements. The most defensible mechanism claim is mediated: GenAI affects product outcomes primarily through changes in process (SRL behaviors), potentially moderated by state:

$$\mathbf{p}_i = \alpha_0 + \alpha_1 \text{GenAI}_i + \alpha_2 \mathbf{X}_i + \eta_i, \tag{15}$$

$$y_i = \beta_0 + \beta_1 \text{GenAI}_i + \beta_2 \mathbf{p}_i + \beta_3 \mathbf{X}_i + \epsilon_i, \tag{16}$$

where $\mathbf{X}_i$ captures task/design covariates. Under this view, neuro/biometric indicators are best used selectively to test hypotheses about how actionable, rubric-anchored feedback influences attention allocation and verification behaviors. This supports a scalable principle: use logs/traces for cohort-wide SRL-aligned evaluation; reserve neuro/biometrics for targeted, hypothesis-driven mechanism studies [15, 16].

4.4 Implications for design and evaluation in higher education

The findings converge on a practical lesson consistent with TKL: GenAI strengthens feedback systems when embedded in criterion-referenced, verifiable workflows that make SRL behaviors observable and consequential [7, 9, 36]. Three actionable components emerge:

1. **Rubric anchoring and observable criteria.** Treat the rubric as a shared language between learner, instructor, and GenAI feedback partner. Require feedback to cite relevant criteria, descriptors, and thresholds to reduce drift and improve auditability [8].
2. **Worked examples and counterexamples.** Provide exemplars that operationalize quality and reduce ambiguity. Pair worked examples with counterexamples to expose typical failure modes and stabilize feedback toward recognized standards [7].
3. **Actionability with verification routines.** Convert feedback into stepwise, checkable actions: "what to change", "why it matters", and "how to check". Verification can be lightweight (checklists, self-explanations) or strict (unit tests, reproducible runs). This design choice is pivotal because it turns GenAI suggestions into assessable learning actions and supports SRL monitoring and reflection [12, 14].

Evaluation should therefore shift from "does it improve grades?" to "does it improve the feedback-driven learning process?" A minimal instrumentation baseline (iterations, latency, retries, verification evidence) is feasible in most higher-education platforms and yields SRL- relevant indicators at scale [13, 16]. Neuro/biometric measures are best framed as complementary mechanism tools in targeted subsamples, integrated with traces rather than treated as standalone evidence of effectiveness.

4.5 Strengths and limitations

This review has several strengths: PRISMA 2020 compliance; explicit moderator coding focused on implementable design features; separation of product, process, and mechanism evidence; conservative handling of multiplicity via one consolidated product effect per study; and dependence-robust sensitivity checks (cluster-robust inference) supporting transparent synthesis decisions [20]. However, limitations warrant cautious interpretation.

First, heterogeneity in tasks, interventions, outcome operationalizations, and reporting practices complicates comparability and may inflate uncertainty. Second, subgroup and meta-regression findings are observational and potentially confounded; causal interpretations should be avoided [32]. Third, despite strict duplicate-resolution

rules, residual dependence across related reports cannot be fully excluded. Fourth, SRL/process measurement remains inconsistent; trace indicators are often platform-specific and not standardized, limiting harmonization [15, 16]. Fifth, risk-of-bias concerns (Figure 5) and incomplete reporting in primary studies constrain inference, and publication-bias diagnostics are underpowered within subgroups; absence of significant asymmetry does not guarantee absence of bias [30].

Automation-assisted screening. Although no record was excluded solely based on automation output, LLM-assisted triage can raise community concerns about silent false negatives. To address this risk, we conducted a false-negative audit by rescreening a random sample of records that the assistant flagged as likely exclusions (audit n=**100**; reinstated to full-text n=**3**; ultimately included n=**1**). While this provides additional reassurance, auditing was necessarily sample-based and cannot completely eliminate the possibility of missed eligible studies.

Single-group evidence and external validity. A non-trivial portion of the corpus ($n = 17$) relied on single-group pre–post designs. These studies often reflect classroom-realistic implementations and therefore strengthen ecological validity, but they are more vulnerable to history/maturation and were not pooled with controlled comparisons. Descriptively, their outcome direction was Descriptively, their outcome direction was **predominantly positive** and was most interpretable when verification artifacts and trace-based process evidence were available, consistent with the review's broader design-and-evaluation claims.

Finally, several presentation constraints matter for interpretability: forest and funnel plots should remain legible at journal column width (e.g., $\geq$300 DPI export, readable axis labels), and study labels should ideally include Author (Year) when space permits to reduce the need for cross-referencing.

4.6 Future research directions

Future work should prioritize: (i) preregistered designs and standardized outcome reporting; (ii) retention and transfer outcomes (beyond immediate performance); (iii) SRL-aligned process instrumentation as a first-class outcome; and (iv) experimental manipulation of design components (rubric anchoring, examples/counterexamples, verification routines) to identify causal mechanisms and boundary conditions [17–19]. For programming and other formally constrained tasks in particular, studies should treat verification as a core design variable (e.g., mandated unit tests, structured debugging prompts, and traceable run-evidence) to directly test whether "fluency trap" risks are mitigated by enforced checking routines.

Neuro/biometric studies should be framed as mechanism-oriented: hypothesis-driven, transparently preprocessed, and explicitly linked to trace-based SRL process outcomes so that state indicators explain *how* and *when* SRL behaviors change in GenAI-mediated feedback environments. Until larger bodies of comparable studies exist, such measures should be interpreted as explanatory complements rather than a basis for general effectiveness claims.

Overall, the evidence supports cautious optimism. GenAI can strengthen formative assessment in higher education, particularly when embedded in explicit, criterion-referenced designs and evaluated using both product outcomes and SRL-aligned

process measures. In TKL terms, the contribution of GenAI is not simply faster feedback, but the potential to reshape the feedback ecology into a learnable, observable cycle of knowledge construction and self-regulated revision.

5 Conclusion

This PRISMA 2020 systematic review and meta-analysis synthesized empirical evidence (2022–2025) on GenAI/LLM-supported feedback in higher education, separating (i) learning-related *product* outcomes, (ii) SRL-aligned *process* outcomes, and (iii) mechanism-oriented evidence from optional neuro/biometric measures. In line with PRISMA 2020 reporting and transparency principles, we documented the full search strategy, screening workflow, and synthesis decisions to support reproducibility and cumulative comparison [20, 21]. Overall, the evidence indicates that GenAI-supported feedback can improve product outcomes on average, while effects vary meaningfully across contexts depending on task demands, feedback design quality, and the measurability of the feedback cycle.

From a synthesis standpoint, the results support a random-effects interpretation in which observed effects can be viewed as

$$y_i = \mu + u_i + \varepsilon_i, \quad u_i \sim \mathcal{N}(0, \tau^2), \tag{17}$$

consistent with standard meta-analytic formulations for heterogeneous educational interventions [24, 25, 32]. This implies that effectiveness is not a fixed constant but a distribution shaped by implementation conditions. Accordingly, the main contribution of this study is not only the pooled estimate of average effectiveness, but a *design-and-evaluation blueprint* that clarifies what must be in place for GenAI to function as a defensible formative-assessment component in higher education. In TKL terms, GenAI is best understood as a *feedback partner* within a learning environment that mediates how students interpret standards, regulate revision decisions, and construct knowledge through iterative comparison between drafts and criteria.

A practical blueprint for implementation.

Impact depends less on adopting GenAI *per se* and more on how it is embedded into criterion-referenced instruction and assessment. Three implementation requirements emerged as consistently actionable and theoretically grounded in the broader feedback and SRL literatures [5, 6, 10, 11]:

1. **Rubric anchoring with observable criteria.** Prompts and outputs should be constrained by explicit rubric dimensions and performance descriptors to improve alignment, reduce drift, and make feedback auditable at the level of standards. This strengthens interpretability and supports feedback uptake in ways consistent with effective formative feedback principles [6, 7].
2. **Examples/counterexamples.** Worked exemplars operationalize quality and reduce ambiguity, while counterexamples make common errors visible and help prevent superficial "polishing" from being mistaken for learning. This aligns with the

view that feedback is most effective when it clarifies what quality looks like and supports learners in calibrating their judgments [7, 9].

3. **Actionability with verification.** Feedback should specify concrete revision steps and corresponding verification routines (e.g., checklists, tests, self-explanations), converting suggestions into assessable actions and supporting SRL cycles of monitoring and reflection [10, 11]. Together, these constraints transform GenAI from a conversational assistant into a criterion-referenced feedback environment that supports uptake, reduces over-reliance, and enables instructors to evaluate whether improvement is attributable to regulated learning activity rather than opaque co-production.

What should be measured (beyond grades).

To avoid conflating improved outputs with improved learning, evaluation should jointly report product and process outcomes. A scalable baseline is achievable via minimal log-based instrumentation that captures SRL-relevant traces (e.g., iterations, feedback-to-revision latency, retries, and evidence of planning/checking), complemented where possible by validated SRL self-report measures. This aligns with learning analytics and SRL measurement arguments that trace data can provide observable, action-level indicators of regulation and engagement when appropriately interpreted [13–15]. Reporting both layers clarifies whether GenAI accelerates the *feedback cycle* (revision and verification routines) and not merely the surface quality of final submissions.

Role of neuro/biometric measures.

Neuro/biometric measures are valuable for mechanism-oriented explanation (e.g., attention allocation, cognitive load), but current evidence supports their use primarily in targeted subsamples rather than as a cohort-wide requirement. The most scalable approach is to treat logs/traces as the default layer for process measurement and add neuro/biometric data selectively for hypothesis-driven mechanism studies that link state indicators to trace-based SRL behaviors [15, 16].

Research agenda.

Future work should prioritize: (i) preregistered, longer-horizon studies assessing retention and transfer; (ii) standardized reporting of trace-based SRL/process metrics across platforms to support cumulative comparison; and (iii) component-level experiments that manipulate rubric anchoring, examples/counterexamples, and verification routines to identify causal boundary conditions and design principles. Open and auditable reporting remains essential: sharing full search strategies, screening logs, extraction codebooks, effect-size datasets, and analysis scripts enables independent verification and cumulative progress [20, 32]. In synthesis practice, routine use of bias-robust procedures (design-appropriate risk-of-bias tools and small-study-effect diagnostics) should remain standard [22, 23, 30].

In sum, GenAI can act as an accelerator of iterative feedback cycles in higher education when embedded in criterion-referenced instructional designs and evaluated

with a dual focus on product performance and SRL-aligned process change, with mechanism measures used strategically to explain *why* and *when* effects emerge [1, 2].

Declarations

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Clinical trial number

Not applicable.

Competing interests

The authors declare that they have no competing interests.

Ethics approval and consent to participate

Not applicable. This study is a systematic review and meta-analysis of published studies and did not involve the collection of primary data from human participants.

Consent for publication

Not applicable.

Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Code availability

Not applicable.

Authors' contributions

R.J.C. and C.B.C. conceptualized the study, developed the PRISMA 2020 study protocol, and designed the search strategy. R.J.C. and C.B.C. independently screened the records, extracted the data, and conducted the risk of bias assessments. A.H.F. served as the third reviewer to adjudicate any disagreements during the screening and extraction phases. R.J.C. conducted the quantitative synthesis and meta-analysis using R. C.B.C. and A.H.F. wrote the main manuscript text and prepared the figures and tables. All authors reviewed, edited, and approved the final manuscript.

Table A1 LLM-assisted triage audit (false-negative check) for screening support.

Audit element	Count
Random audit sample (LLM-flagged exclusions)	100
Judged potentially eligible upon human review	3
Ultimately included after full-text assessment	1

Appendix A Search strategies and audit table

A.1 Search audit trail

Table A2 reports the audit trail for each pre-specified query variant, including the database, query ID, fields searched, limits applied, execution date, and records exported. This audit trail supports reproducibility and complements the PRISMA flow diagram (Figure 3) in line with PRISMA 2020 reporting guidance [20, 21]. All database queries were executed on **16 December 2025** (Europe/Madrid time). Records were exported with full bibliographic metadata and abstracts (and references when available), then merged and deduplicated in Zotero prior to screening.

Note on overlap across query variants. The four query variants (Core; Core+SRL; Core+Logs; Core+Neuro) were designed to maximize recall for distinct construct families. Because variants may retrieve overlapping records, we maintained an explicit export log and removed overlaps during merging/deduplication. Counts in Table A2 reflect the *exported records attributed to each query variant after within-database overlap handling*, and therefore sum to the total records exported from each database reported in the PRISMA flow (WoS n=520; Scopus n=480). Reviews were retrieved to maximize recall and support snowballing, but were excluded from quantitative pooling and retained only for context when relevant (Section 2.2).

Automation transparency note (screening support). In addition to the database audit trail, we retained an automation audit log for the LLM-assisted *pre-tagging/triage* used to prioritize human screening (see Section 2.5 and Section 2.10). No record was excluded solely on the basis of automation output; all eligibility decisions were made by human reviewers. To address concerns about silent false negatives, we **performed** a formal false-negative audit of LLM-flagged exclusions; details are reported in Appendix Table A1.

A.2 Web of Science Core Collection (TS=Topic)

The following strings were executed in Web of Science Core Collection using **TS (Topic)**, which searches *title, abstract, author keywords, and Keywords Plus*. Filters were applied for publication years (2022–2025), document types (Article/Review/Proceedings Paper), and language (English; Spanish when applicable). For reproducibility, the Web of Science search-history export (including index selection, timespan settings, and interface-level filters) was retained as part of the audit materials.

```
A) Core (WoS / TS)
```

Table A2 Search strategy audit trail: Yield comparison across databases. Counts represent unique records per query variant prior to global deduplication.

		Database Yield (n)		
ID	**Query Scope / Terms**	**WoS**[a]	**Scopus**[b]	**Total**
A	Core (Baseline)	230	210	**440**
B	Core + SRL	140	150	**290**
C	Core + Logs	95	85	**180**
D	Core + Neuro	55	35	**90**
	Total Exported	**520**	**480**	**1,000**

Note: Date of search: 16/12/2025. Limits: 2022–2025, EN/ES.
[a] Field: TS (Topic). [b] Field: TITLE-ABS-KEY.

```
TS=(
  ("chatgpt" OR "generative ai" OR "generative artificial intelligence" OR genai OR
   "large language model*" OR llm* OR "foundation model*" OR
   "gpt-3" OR "gpt-3.5" OR "gpt-4" OR "language model*" OR
   "ai writing assistant*" OR "conversational agent*" OR chatbot*)
  AND
  (feedback OR "automated feedback" OR "formative feedback" OR "written feedback" OR
   "feedback generation" OR "automated assessment" OR "automated grading" OR
   rubric* OR "intelligent tutoring" OR tutor*)
  AND
  ("higher education" OR universit* OR "tertiary education" OR college OR
   undergraduate* OR postgraduate* OR "graduate student*" OR "university student*")
)

B) Core + SRL (WoS / TS)
TS=(
  (("chatgpt" OR "generative ai" OR genai OR "large language model*" OR llm* OR
     "foundation model*" OR "gpt-4" OR chatbot*)
   AND (feedback OR "automated feedback" OR "automated assessment" OR rubric* OR
        tutor*)
   AND ("higher education" OR universit* OR undergraduate* OR postgraduate* OR
        "university student*"))
  AND
  ("self-regulated learning" OR SRL OR self-regulat* OR metacognit* OR
   "goal setting" OR "self-monitor*" OR "learning strateg*" OR "help seeking")
)

C) Core + Logs / Learning Analytics (WoS / TS)
TS=(
  (("chatgpt" OR "generative ai" OR genai OR "large language model*" OR llm* OR
     "foundation model*" OR "gpt-4" OR chatbot*)
   AND (feedback OR "automated feedback" OR "automated assessment" OR rubric* OR
```

```
        tutor*)
   AND ("higher education" OR universit* OR undergraduate* OR postgraduate* OR
        "university student*"))
  AND
  ("learning analytic*" OR log* OR "trace data" OR clickstream OR "time on task" OR
   "interaction data" OR "revision history" OR "version control" OR commit* OR
   "submission trace*" OR telemetry)
)

D) Core + Neuro/Biometric (WoS / TS)
TS=(
  (("chatgpt" OR "generative ai" OR genai OR "large language model*" OR llm* OR
     "foundation model*" OR "gpt-4" OR chatbot*)
   AND (feedback OR "automated feedback" OR "automated assessment" OR rubric* OR
        tutor*)
   AND ("higher education" OR universit* OR undergraduate* OR postgraduate* OR
        "university student*"))
  AND
  (EEG OR electroencephalograph* OR ERP OR "event-related potential*" OR
   "eye tracking" OR eyetrack* OR pupillometr* OR fNIRS OR
   "cognitive load" OR workload OR attent* OR "physiological measure*")
)
```

A.3 Scopus (TITLE-ABS-KEY)

The following strings were executed in Scopus using **TITLE-ABS-KEY**. Filters were applied for publication years (2022–2025), document types (Article/Review/Conference paper), and language (English; Spanish when applicable). For auditability, we retained the Scopus query history and export parameters (fields exported, execution timestamp, and any interface-level filters) as part of the audit package.

```
A) Core (Scopus / TITLE-ABS-KEY)
TITLE-ABS-KEY(
  ("chatgpt" OR "generative ai" OR "generative artificial intelligence" OR genai OR
   "large language model*" OR llm* OR "foundation model*" OR
   "gpt-3" OR "gpt-3.5" OR "gpt-4" OR "language model*" OR
   "ai writing assistant*" OR "conversational agent*" OR chatbot*)
  AND
  (feedback OR "automated feedback" OR "formative feedback" OR "written feedback" OR
   "feedback generation" OR "automated assessment" OR "automated grading" OR
   rubric* OR "intelligent tutoring" OR tutor*)
  AND
  ("higher education" OR universit* OR "tertiary education" OR college OR
   undergraduate* OR postgraduate* OR "graduate student*" OR "university student*")
)

B) Core + SRL (Scopus / TITLE-ABS-KEY)
TITLE-ABS-KEY(
  (("chatgpt" OR "generative ai" OR genai OR "large language model*" OR llm* OR
```

```
      "foundation model*" OR "gpt-4" OR chatbot*)
    AND (feedback OR "automated feedback" OR "automated assessment" OR rubric* OR
         tutor*)
    AND ("higher education" OR universit* OR undergraduate* OR postgraduate* OR
         "university student*"))
  AND
  ("self-regulated learning" OR SRL OR self-regulat* OR metacognit* OR
   "goal setting" OR "self-monitor*" OR "learning strateg*" OR "help seeking")
)

C) Core + Logs (Scopus / TITLE-ABS-KEY)
TITLE-ABS-KEY(
  (("chatgpt" OR "generative ai" OR genai OR "large language model*" OR llm* OR
      "foundation model*" OR "gpt-4" OR chatbot*)
    AND (feedback OR "automated feedback" OR "automated assessment" OR rubric* OR
         tutor*)
    AND ("higher education" OR universit* OR undergraduate* OR postgraduate* OR
         "university student*"))
  AND
  ("learning analytic*" OR log* OR "trace data" OR clickstream OR "time on task" OR
   "interaction data" OR "revision history" OR "version control" OR commit* OR
   "submission trace*" OR telemetry)
)

D) Core + Neuro (Scopus / TITLE-ABS-KEY)
TITLE-ABS-KEY(
  (("chatgpt" OR "generative ai" OR genai OR "large language model*" OR llm* OR
      "foundation model*" OR "gpt-4" OR chatbot*)
    AND (feedback OR "automated feedback" OR "automated assessment" OR rubric* OR
         tutor*)
    AND ("higher education" OR universit* OR undergraduate* OR postgraduate* OR
         "university student*"))
  AND
  (EEG OR electroencephalograph* OR ERP OR "event-related potential*" OR
   "eye tracking" OR eyetrack* OR pupillometr* OR fNIRS OR
   "cognitive load" OR workload OR attent* OR "physiological measure*")
)
```

A.4 Deduplication and conference-versus-journal audit

To ensure traceability when multiple records described the same underlying study (e.g., a conference paper later extended into a journal article), we maintained an audit log linking related records and applying strict decision rules. These rules preserve independence in the quantitative synthesis and support auditable claims about evidence provenance.

- **Same underlying study (duplicate report):** If conference and journal versions reported the *same sample* (or fully overlapping cohorts), the *same intervention window*, and *overlapping outcomes*, we retained the most complete/definitive report

Table A3 Audit log of provenance decisions for related records (conference vs. journal). This log ensures non-redundant data extraction while preserving lineage.

ID	Conference Record	Journal Record	Decision	Rationale for Selection
S_01	Author A (2022), *LAK Proc.*	Author A (2023), *Comput. Educ.*	**Retain Journal**	Identical cohort and intervention. Journal version includes full psychometric validation and granular metrics.
S_02	Author B (2024), *AIED Proc.*	—	**Retain Conf.**	No journal counterpart identified. Source is peer-reviewed and contains extractable statistical data.
S_03	Author C (2022), *Euro-Call*	Author C (2024), *BJET*	**Include Both**	Distinct datasets: Journal extends follow-up by 6 months and reports different dependent variables (linked as *Family ID*).

Note: Priority rule: Completeness $\succ$ Recency. Overlapping samples are treated as a single unit of analysis.

(typically the journal article) and recorded the alternative as a related record (not double-counted).

- **Extended study (not a duplicate):** If a later report added a *new sample*, *additional cohorts*, *longer follow-up*, or clearly *non-overlapping outcomes*, it was treated as a distinct study while still linking records for provenance.
- **Effect-size extraction rule:** When records were linked, effect sizes were extracted from the retained definitive report. If a secondary report contained unique extractable statistics not present in the definitive report, they were used only when they referred to non-overlapping outcomes/time points and were explicitly documented to avoid duplication.

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